

Mu-Te Lau

Graduate Researcher Specialized in Quantum Circuit Compilation

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Education

Northwestern University

Ph.D. in Computer Science (Advisor: Nikos Hardavellas)

Evanston, IL, USA

Sep. 2025–Jun. 2030 (Expected)

National Taiwan University

M.S. in Electrical Engineering (Advisor: Chung-Yang (Ric) Huang, GPA: 3.94/4.30)

Taipei, Taiwan

Sep. 2022–Jun. 2024

- **Received 2022 GIEE Scholarship for Outstanding Academic Performance (Top 8% GPA in 236 students)**
- **Completed the Quantum Computation and Quantum Information Program** organized by Dept. of Physics, NTU

National Taiwan University

B.S. in Electrical Engineering (GPA: 3.86/4.30; GPA since junior: 3.95/4.30)

Taipei, Taiwan

Sep. 2017–Jun. 2022

Research Experience

Northwestern University

Graduate Research Assistant (Advisor: Nikos Hardavellas; in collaboration with Kaitlin N. Smith)

Evanston, IL, USA

Sep. 2025–Present

- Developing hardware-aware compilation techniques for quantum Hamiltonian simulation
- Studying stabilizer theories for CV-DV quantum computer compilations
- Architecting an MLIR-based quantum compiler frameworks to support hybrid quantum architectures

Design Verification Lab, National Taiwan University

Research Assistant; later promoted to Research Associate (Advisor: Chung-Yang (Ric) Huang)

Taipei, Taiwan

Sep. 2022–Feb. 2025

- Initiated and led Qsyn as primary developer, growing it into an open-source quantum circuit synthesis framework with 190+ GitHub stars
- Designed Qsyn as a platform for users to compose and explore alternative quantum compilation strategies
- Reimplemented and advanced tableau-based quantum circuit optimization methods to reduce 2Q-count overhead

Publications

SEQC: Stratify-Elaborate Quantum Compilation Towards Modular Hybrid Architectures

Northwestern University, IL, USA

Mingyoung Jessica Jeng, Nikola Vuk Maruszewski, **Mu-Te Lau**, Connor Selna, Michael Gavrincea, Kaitlin N. Smith, Nikos Hardavellas

Sep. 2026, IEEE QCE'26

- **Achieved 9.3–32.3% average circuit-fidelity improvement and 1.34–3.27× faster compilation over a chiplet-unaware Qiskit baseline**
- Parallelization-friendly compilation for modular architectures with novel placement, routing, and optimization methods

A Lazy Resynthesis Approach for Simultaneous T Gate and Two-Qubit Gate Optimization of Quantum Circuits

National Taiwan University, Taiwan

Mu-Te Lau, Hsiang-Chun Yang, Hsin-Yu Chen, Chung-Yang (Ric) Huang

Sep. 2025, IEEE QCE'25

- **Presented at IEEE Quantum Week (QCE) 2025**
- **Reduced 2Q-count overhead by 54.8% for tableau-based quantum circuit optimization while achieving 1.81× speedup**

Multi-Objective Quantum Circuit Optimization by Combining Tableau-Based and ZX-Diagram-Based Techniques

National Taiwan University, Taiwan

Mu-Te Lau (Advisor: Chung-Yang (Ric) Huang)

Jul. 2024, Master's Thesis

- **Proposed a hybrid QCO flow for Clifford+T circuits that give a 29.4% improvement in 2Q-counts over purely tableau-based flows**

Qsyn: A Developer-Friendly Quantum Circuit Synthesis Framework for NISQ Era and Beyond

National Taiwan University, Taiwan

Mu-Te Lau, Chin-Yi Cheng, Cheng-Hua Lu, Chung-Yang (Ric) Huang (Corresponding Author), et al.

Apr. 2024, arXiv Preprint

- A fast and modular framework for quantum circuit synthesis, offering a flexible and extendable CLI for pass composition
- **Poster presented at IEEE QCE'24 and 6th IWQC**

Teaching Experiences

Special Topics on Quantum Design Automation

National Taiwan University, Taiwan

Head of Teaching Assistant, Graduate Institute of Electrical Engineering

2023 Fall

- Instructors: Profs. Chung-Yang (Ric) Huang, Jie-Hong (Roland) Jiang, James Chien-Mo Li, Shih-Hao Hung
- **Gave a TA lecture on ZX-calculus-based Quantum Circuit Optimization**
- Designed and graded assignments and final exams

Quantum Information and Computation

National Taiwan University, Taiwan

Head of Teaching Assistant, Graduate Institute of Electrical Engineering

2023 and 2024 Spring

- Instructor: Prof. Hao-Chung Cheng
- Designed and graded assignments and exams

Web Programming

National Taiwan University, Taiwan

Teaching Assistant, Department of Electrical Engineering

2022 and 2023 Fall

- Instructor: Prof. Chung-Yang (Ric) Huang
- Graded term projects, designed programming assignments, and maintained the course website

Project Experiences

Qsyn | arXiv  |  190+ ★

National Taiwan University, Taiwan

Quantum Computing; Modern C++; Docker

2022 Fall–Now

- **Reimplemented and improved QCO algorithms to assess for scalable, high-performance quantum circuit synthesis**
- Implemented a flexible command-line interface to combine QCO algorithms flexibly
- Coordinated refactorings to core data structures to ensure code quality and flexibility
- Guided new team members on their contributions and taught them good coding practices

ZX-Diagrams as Intermediate Representation for Lattice Surgery Compilation

National Taiwan University, Taiwan

Survey, C++

2022 Spring–2023 Summer

- Term projects of the courses *Fault-Tolerant Computing* and *Quantum Information and Computation*
- **Selected to be Exemplar Presentation Videos in the 2022 Quantum Information and Computation Course**
- Compiled Fault-Tolerant Quantum Circuit to Lattice Surgery with ZX-calculus-based methods
- Achieved compact compilation results for quantum circuits with a small number of qubits

Leadership Experiences

Band Leader; Chair Euphonium Player; Social Media Editor

Taipei, Taiwan

National Taiwan University Wind Band

Aug. 2019–Aug. 2024

- Coordinated, as the band leader, the band's rehearsals and performances and solved administrative difficulties during the COVID pandemic
- Promoted, as the social media editor, the band's events by garnering over 169.7K in reach and growing Instagram followers by 43%

Server & Network Administrator

Taipei, Taiwan

Design Verification Lab, National Taiwan University

Feb. 2022–Feb. 2025

- Maintained the lab servers and pertinent hardware such as routers, NAS, and firewalls
- Built comprehensive documentation for future administrators

Skills

Programming	Modern C++, Shell, Python, JavaScript, Rust
Quantum Computing Tools	Qiskit, PyZX, Feynman
Web Development	JS/React, Next.js, Docker, MongoDB
Languages	Mandarin (Native), English (Proficient), Japanese (Basic), German (Basic)